Thuan Nguyen

Homework 3

1. Manual Gradient Derivation

The Skip Gram with Negative Sampling (SGNS) model learns word embeddings by bringing target and positive context vectors closer together while separating target vectors from negative samples.

Let u_t represent the target embedding, v_c represent the positive context embedding, and v_nk represent the kth negative context embedding. The sigmoid function is represented by σ(x).

The batch mean SGNS loss is calculated by taking the negative average of the log sigmoid score for each positive context word and the log sigmoid scores for all negative samples. In this formula, B represents the batch size, and K represents the number of negative samples for each positive pair.

The gradient for a target embedding includes two parts. The first part comes from the positive context word and encourages the target and positive context embeddings to become more closely related. The second part comes from the negative samples and helps the model distinguish unrelated words from positive context words. The gradient is divided by B to account for averaging the loss across the batch.

For the positive context embedding, the gradient is the sigmoid of the target and context dot product minus one, multiplied by the target embedding and divided by the batch size. For each negative context embedding, the gradient is the sigmoid of the target and negative context dot product, multiplied by the target embedding and divided by the batch size.

The implementation calculates these gradients manually inside torch.no_grad() and accumulates the contributions when multiple examples reference the same embedding index.

2. Gradient Verification

The word2vec_verification.py script compares the manually calculated gradients with gradients generated by PyTorch autograd. The test uses torch.allclose() with a tolerance of 10^{-5}.

The verification completed successfully, confirming that the manual gradients match the autograd results within the specified numerical tolerance for the tested inputs.

3. Hyperparameter Analysis

The number of negative samples, K, affects training cost and the information used to distinguish real context words from noise. Increasing K provides more negative examples for each positive pair, but also increases computation. Using too few negative samples may provide less information for learning useful word representations.

The embedding dimension, d, determines how many values are used to represent each word. A larger dimension can capture more relationships between words, but requires more memory and computation and may overfit small datasets. A smaller dimension is more efficient but may not capture enough information to represent complex relationships.

The learning rate also affects optimization. A learning rate that is too high can make training unstable, while a rate that is too low can slow convergence. In my experiments, the average SGNS loss decreased over ten epochs, indicating that the model was learning to optimize its objective.

4. Conclusion

This assignment involved implementing a custom SGNS data pipeline, a vectorized forward pass, and an analytical manual backpropagation method. Gradient verification confirmed that the manually calculated gradients matched the PyTorch autograd results for the tested inputs. The training results also showed decreasing average loss across ten epochs, demonstrating that the optimization process was progressing.